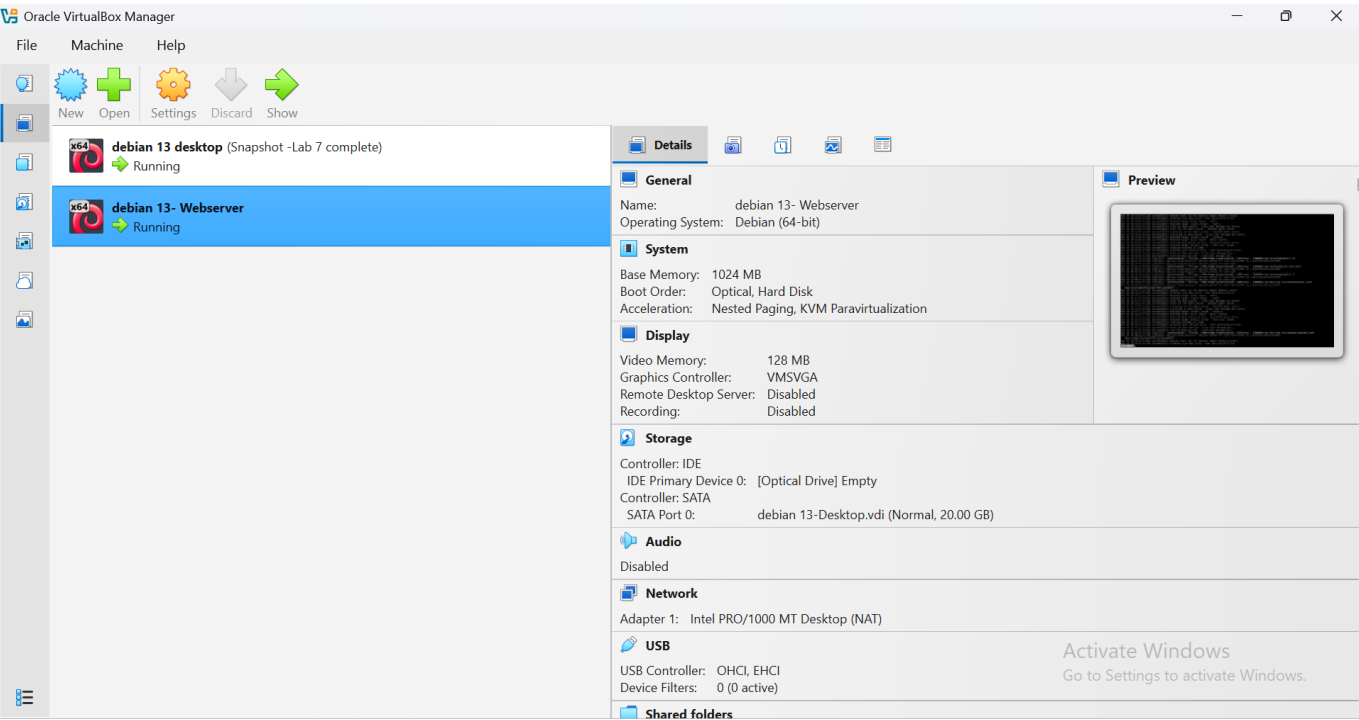


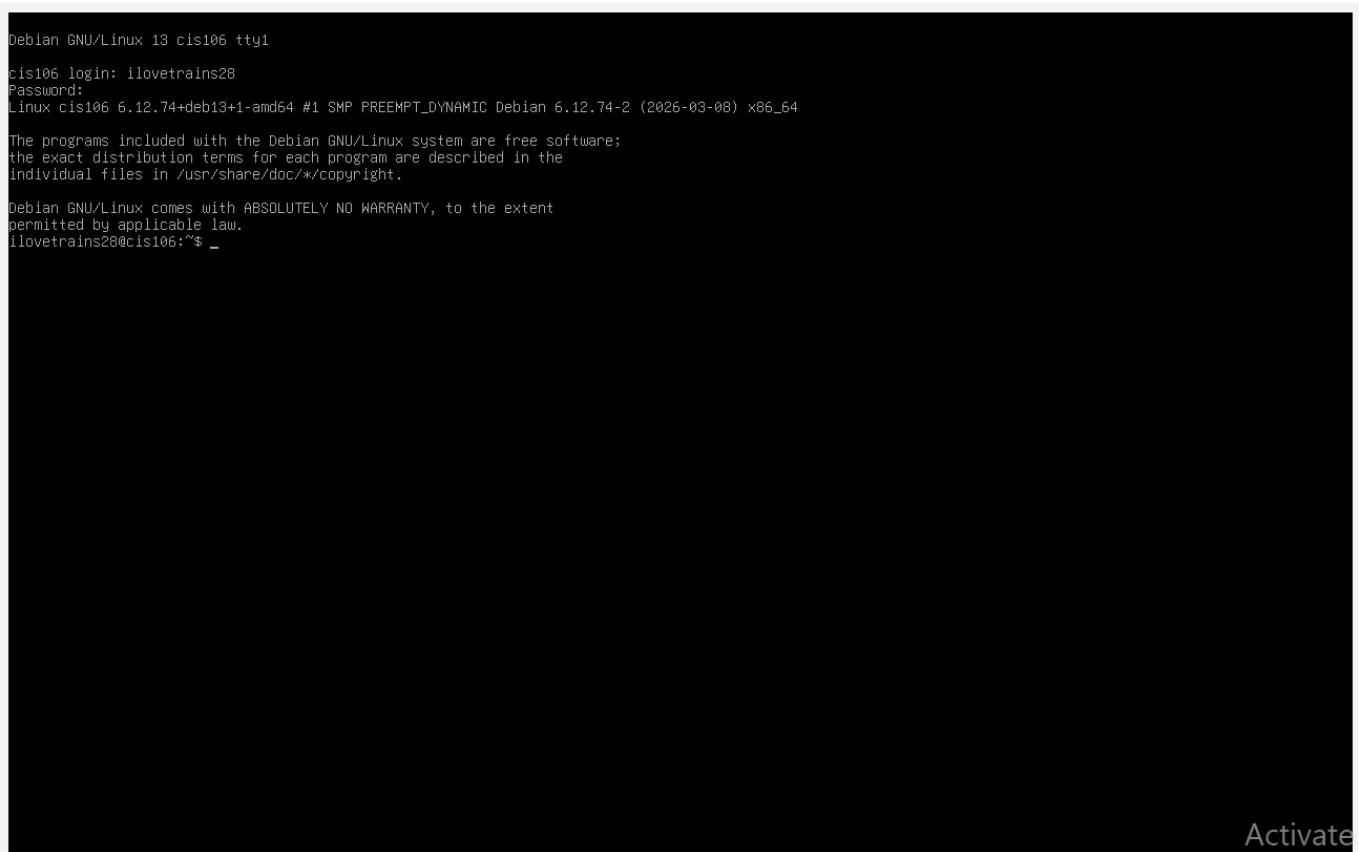
Deliverable 2

Questions

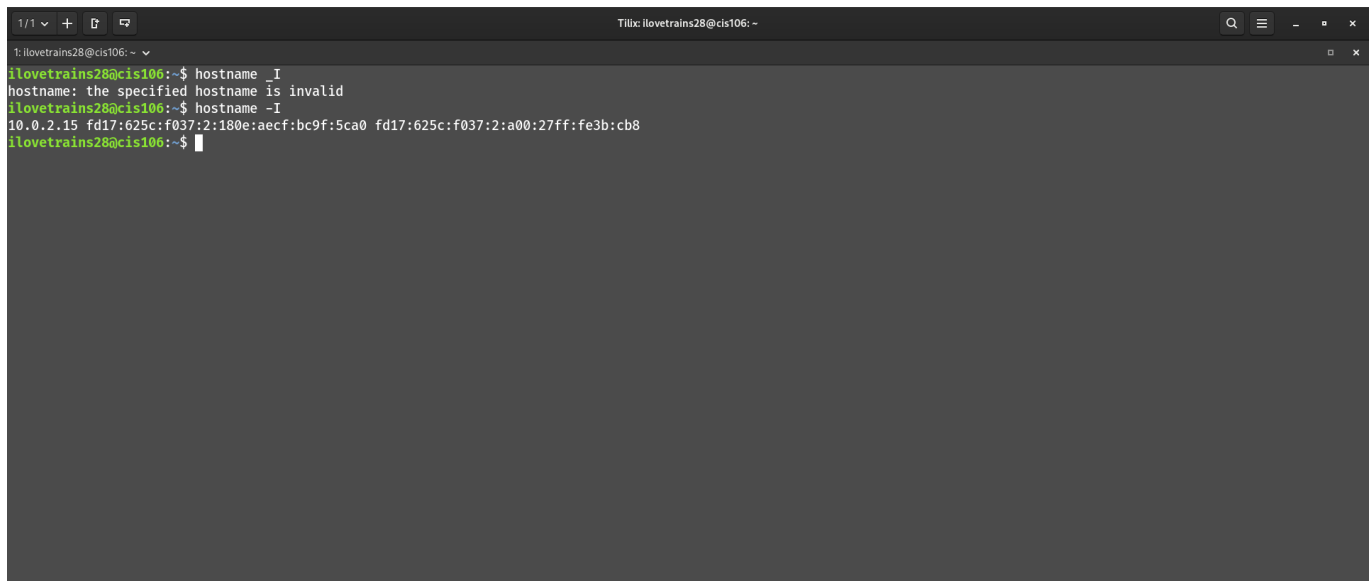
1. What are the server hardware specifications (virtual machine settings)?



2. What is the Debian Login Screen?



3. What is the IP address of your Debian Server Virtual Machine?



```
1/1 v + [ ] [ ]
Tilix: ilovetrains28@cis106: ~
ilovetrains28@cis106:~$ hostname -I
hostname: the specified hostname is invalid
ilovetrains28@cis106:~$ hostname -I
10.0.2.15 fd17:625c:f037:2:180e:aecf:bc9f:5ca0 fd17:625c:f037:2:a00:27ff:fe3b:cb8
ilovetrains28@cis106:~$
```

Hostname -I

4.

Command name

- **Description:** Est pariatur sint fugiat mollit ea est veniam proident nulla laboris excepteur.
- **Formula/Syntax:** `sudo + ufw + argument`
- **Examples:**
 - How do you check if the Firewall is running?
 - By using the command `sudo + ufw + status`
 - Checks to see if ufw is enabled.
 - How do you disable the Firewall?
 - By using the command `sudo + ufw + disable`
 - Fully disables the firewall service in your system.
 - How do you add Apache to the Firewall?
 - By using the command `sudo + ufw + allow + "Apache Full"`
 - Allows both HTTP and HTTPS traffic on the server.

5. What different commands do we use to work with Apache?

1. What is the command you use to check if Apache is running?

- The command is: `systemctl + status + apache2`

- Checks that Apache is enabled and running.

```
Debian GNU/Linux 13 cis106 tty1
cis106 login: ilovetrains28
Password:
Linux cis106 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
ilovetrains28@cis106:~$ systemctl status apache2
• apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-21 13:20:08 EDT; 8min ago
   Invocation: bf3bce181120479890e0b9f0bc9f2d08
   Docs: https://httpd.apache.org/docs/2.4/
   Process: 645 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 677 (apache2)
   Tasks: 55 (limit: 1093)
   Memory: 8.1M (peak: 8.4M)
   CPU: 133ms
   CGroup: /system.slice/apache2.service
           └─677 /usr/sbin/apache2 -k start
             └─681 /usr/sbin/apache2 -k start
               └─685 /usr/sbin/apache2 -k start

Warning: some journal files were not opened due to insufficient permissions.
ilovetrains28@cis106:~$
```

Activate V

2. What is the command you use to stop Apache?

- The command is: **sudo + systemctl + stop + apache2**
- Stops apache

```
Debian GNU/Linux 13 cis106 tty1
cis106 login: ilovetrains28
Password:
Linux cis106 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
ilovetrains28@cis106:~$ systemctl status apache2
• apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-21 13:20:08 EDT; 8min ago
   Invocation: bf3bce181120479890e0b9f0bc9f2d08
   Docs: https://httpd.apache.org/docs/2.4/
   Process: 645 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 677 (apache2)
   Tasks: 55 (limit: 1093)
   Memory: 8.1M (peak: 8.4M)
   CPU: 133ms
   CGroup: /system.slice/apache2.service
           └─677 /usr/sbin/apache2 -k start
             └─681 /usr/sbin/apache2 -k start
               └─685 /usr/sbin/apache2 -k start

Warning: some journal files were not opened due to insufficient permissions.
ilovetrains28@cis106:~$ sudo systemctl stop apache2
[sudo] password for ilovetrains28:
ilovetrains28@cis106:~$
```

Activate V

3. What is the command you use to restart Apache?

- The command is: **sudo + systemctl + restart + apache2**

- Reloads Apache

```
Debian GNU/Linux 13 cis106 tty1
cis106 login: ilovetrains28
Password:
Linux cis106 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
ilovetrains28@cis106:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-21 13:20:08 EDT; 8min ago
  Invocation: bf3bce181120479890e0b9f0bc9f2d08
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 645 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 677 (apache2)
       Tasks: 55 (limit: 1093)
      Memory: 8.1M (peak: 8.4M)
         CPU: 133ms
    CGroup: /system.slice/apache2.service
            └─677 /usr/sbin/apache2 -k start
              └─681 /usr/sbin/apache2 -k start
                └─685 /usr/sbin/apache2 -k start

Warning: some journal files were not opened due to insufficient permissions.
ilovetrains28@cis106:~$ sudo systemctl stop apache2
[sudo] password for ilovetrains28:
ilovetrains28@cis106:~$ sudo systemctl start apache2
ilovetrains28@cis106:~$ sudo systemctl restart apache2
ilovetrains28@cis106:~$
```

Activate

4. What is the command used to test Apache configuration?

- The command is: **sudo + apachectl + configtest**
- Tests Apache's configuration.

```
Debian GNU/Linux 13 cis106 tty1
cis106 login: ilovetrains28
Password:
Linux cis106 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
ilovetrains28@cis106:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-21 13:20:08 EDT; 8min ago
  Invocation: bf3bce181120479890e0b9f0bc9f2d08
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 645 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 677 (apache2)
       Tasks: 55 (limit: 1093)
      Memory: 8.1M (peak: 8.4M)
         CPU: 133ms
    CGroup: /system.slice/apache2.service
            └─677 /usr/sbin/apache2 -k start
              └─681 /usr/sbin/apache2 -k start
                └─685 /usr/sbin/apache2 -k start

Warning: some journal files were not opened due to insufficient permissions.
ilovetrains28@cis106:~$ sudo systemctl stop apache2
[sudo] password for ilovetrains28:
ilovetrains28@cis106:~$ sudo systemctl start apache2
ilovetrains28@cis106:~$ sudo systemctl restart apache2
ilovetrains28@cis106:~$ sudo apachectl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress th
is message
Syntax OK
ilovetrains28@cis106:~$
```

Activate

5. What is the command used to check the installed version of Apache?

- The command is: **/usr/sbin/apache2 + -v**

- Checks current Apache version.

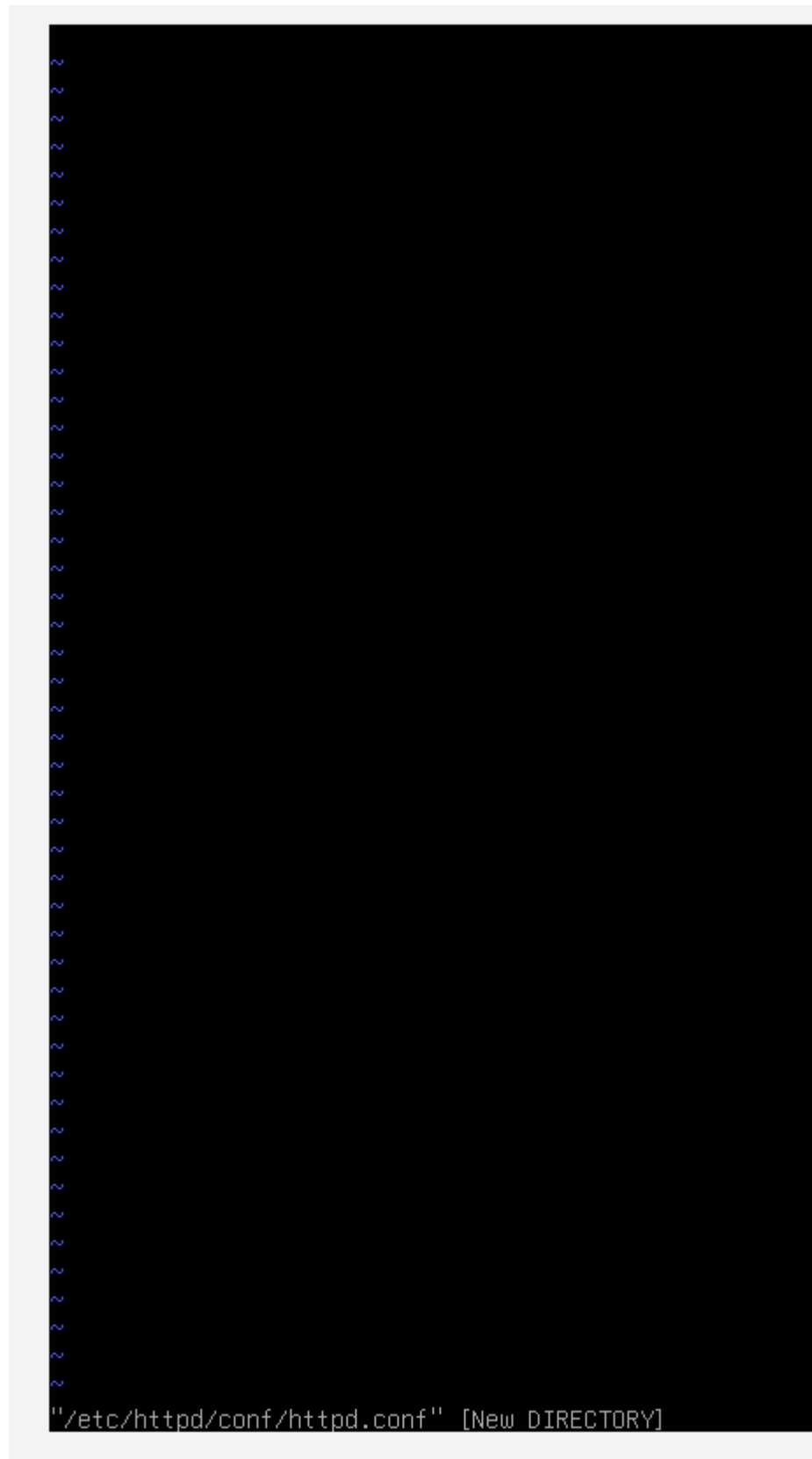
```
ilovetrains28@cis106:~$ apache2 --version
-bash: apache2: command not found
ilovetrains28@cis106:~$ httpd -v
-bash: httpd: command not found
ilovetrains28@cis106:~$ apachectl -v
-bash: apachectl: command not found
ilovetrains28@cis106:~$ /usr/sbin/apache2 -v
Server version: Apache/2.4.66 (Debian)
Server built: 2026-03-01T13:26:45
ilovetrains28@cis106:~$
```

Activate \

6. What are some common configuration files for Apache?

- The command is: `sudo + vim + /etc/httpd/conf/httpd.conf`

- Opens the web server's main configuration file in Vim.

A screenshot of a Vim editor window. The window has a black background with blue text. On the left side, there is a vertical column of blue tilde (~) characters. At the bottom of the window, the text `"/etc/httpd/conf/httpd.conf" [New DIRECTORY]` is displayed in blue.

- The command is: `sudo + vim + /etc/apache2/ports.conf`

- Defines what ports Apache listens on (e.g., 80, 443).

```
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 80

<IfModule ssl_module>
    Listen 443
</IfModule>

<IfModule mod_gnutls.c>
    Listen 443
</IfModule>

"/etc/apache2/ports.conf" 13L, 274B
```

- The command is: `sudo + vim + /etc/apache2/apache2.conf`
- The file that controls global server behavior.

```
# This is the main apache server configuration file. It contains the
# configuration directives that give the server its instructions.
# See http://httpd.apache.org/docs/2.4/ for detailed information about
# the directives and /usr/share/doc/apache2/README.Debian about Debian specific
# hints.
#
#
# Summary of how the Apache 2 configuration works in Debian:
# The Apache 2 web server configuration in Debian is quite different to
# upstream's suggested way to configure the web server. This is because Debian's
# default apache2 installation attempts to make adding and removing modules,
# virtual hosts, and extra configuration directives as flexible as possible, in
# order to make automating the changes and administering the server as easy as
# possible.
#
# It is split into several files forming the configuration hierarchy outlined
# below, all located in the /etc/apache2/ directory:
#
#
#     /etc/apache2/
#     |-- apache2.conf
#     |   |-- ports.conf
#     |-- mods-enabled
#     |   |-- *.load
#     |   |-- *.conf
#     |-- conf-enabled
#     |   |-- *.conf
#     |-- sites-enabled
#         |-- *.conf
#
#
# * apache2.conf is the main configuration file (this file). It puts the pieces
# together by including all remaining configuration files when starting up the
# web server.
#
# * ports.conf is always included from the main configuration file. It is
# supposed to determine listening ports for incoming connections which can be
# customized anytime.
#
# * Configuration files in the mods-enabled/, conf-enabled/ and sites-enabled/
# directories contain particular configuration snippets which manage modules,
# global configuration fragments, or virtual host configurations,
# respectively.
#
# They are activated by symlinking available configuration files from their
# respective *available/ counterparts. These should be managed by using our
# helpers a2enmod/a2dismod, a2ensite/a2dissite and a2enconf/a2disconf. See
# their respective man pages for detailed information.
#
# * The binary is called apache2. Due to the use of environment variables, in
# /etc/apache2/apache2.conf" 225L, 717B8
```

7. Where does Apache store logs?

- The command is: `sudo + tail + -20 + /var/log/apache2/access.log`

- To view recent log entries.

```
Debian GNU/Linux 13 cis106 tty1
cis106 login: ilovetrains28
Password:
Linux cis106 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
ilovetrains28@cis106:~$ sudo tail -20 /var/log/apache2/access.log
[sudo] password for ilovetrains28:
::1 - - [02/Apr/2026:17:33:36 -0400] "GET / HTTP/1.1" 200 11014 "-" "wget/1.25.0"
127.0.0.1 - - [02/Apr/2026:17:33:56 -0400] "GET / HTTP/1.1" 200 11014 "-" "wget/1.25.0"
192.168.1.1 - - [09/Apr/2026:14:15:43 -0400] "GET / HTTP/1.1" 200 10958 "-" "-"
10.0.2.2 - - [16/Apr/2026:17:15:14 -0400] "GET /portfolio HTTP/1.1" 404 530 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0"
10.0.2.2 - - [16/Apr/2026:17:15:14 -0400] "GET /favicon.ico HTTP/1.1" 404 529 "http://127.0.0.1:8080/portfolio" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0"
10.0.2.2 - - [16/Apr/2026:17:15:22 -0400] "GET / HTTP/1.1" 200 3383 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0"
10.0.2.2 - - [16/Apr/2026:17:15:22 -0400] "GET /icons/openlogo-75.png HTTP/1.1" 200 6040 "http://127.0.0.1:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0"
10.0.2.2 - - [16/Apr/2026:17:16:13 -0400] "-" 408 0 "-" "-"
10.0.2.2 - - [16/Apr/2026:17:21:22 -0400] "GET /portfolio HTTP/1.1" 301 626 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0"
10.0.2.2 - - [16/Apr/2026:17:21:22 -0400] "GET /portfolio/ HTTP/1.1" 200 3072 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0"
10.0.2.2 - - [16/Apr/2026:17:22:13 -0400] "-" 408 0 "-" "-"
ilovetrains28@cis106:~$
```

Activate

8. What are some basic commands we can use to review logs?

- The command is: `sudo + journalctl + -f`
- Follows logs in real-time.

```
ilovetrains28@cis106:~$ sudo journalctl -f
Apr 30 15:59:10 cis106 systemd[824]: Started mpris-proxy.service - Bluetooth mpris proxy.
Apr 30 15:59:10 cis106 systemd[824]: Reached target default.target - Main User Target.
Apr 30 15:59:10 cis106 systemd[824]: Startup finished in 149ms.
Apr 30 15:59:10 cis106 systemd[824]: Created slice session.slice - User Core Session Slice.
Apr 30 15:59:10 cis106 systemd[824]: Starting dbus.service - D-Bus User Message Bus...
Apr 30 15:59:10 cis106 systemd[824]: Started dbus.service - D-Bus User Message Bus.
Apr 30 15:59:32 cis106 systemd-timesyncd[302]: Contacted time server 45.56.66.53:123 (2.debian.pool.ntp.org).
Apr 30 15:59:32 cis106 systemd-timesyncd[302]: Initial clock synchronization to Thu 2026-04-30 15:59:32.013318 EDT.
Apr 30 16:04:41 cis106 sudo[858]: ilovetrains28 : TTY=ttty1 ; PWD=/home/ilovetrains28 ; USER=root ; COMMAND=/usr/bin/journalctl -f
Apr 30 16:04:41 cis106 sudo[858]: pam_unix(sudo:session): session opened for user root(uid=0) by ilovetrains28(uid=1000)
```

Activate

Go to Settings

- The command is: `journalctl + -p + err`

- Filters for error-level messages.

```
ilovetrains28@cis106:~$ journalctl -p err
Hint: You are currently not seeing messages from other users and the system.
Users in groups 'adm', 'systemd-journal' can see all messages.
Pass -q to turn off this notice.
Apr 16 14:59:44 cis106 sudo[968]: ilovetrains28 : user NOT in sudoers ; TTY=ttty1 ; PWD=/home/ilovetrains28 ; USER=root ; COMMAND=/usr/bin/apt install git
Apr 16 15:13:10 cis106 sudo[1165]: ilovetrains28 : user NOT in sudoers ; TTY=pts/0 ; PWD=/var/www/html ; USER=root ; COMMAND=/usr/bin/chown ./ ilovetrains28
Apr 16 15:15:06 cis106 sudo[1172]: ilovetrains28 : user NOT in sudoers ; TTY=pts/0 ; PWD=/var/www/html ; USER=root ; COMMAND=/usr/sbin/usermod -ag sudo ilovetrains28
Apr 16 15:22:34 cis106 sudo[1320]: ilovetrains28 : user NOT in sudoers ; TTY=pts/0 ; PWD=/var/www/html ; USER=root ; COMMAND=/usr/bin/chown ./ ilovetrains28
lines 1-4/4 (END)
```

The command is: `journalctl + --since + "1 hour ago"`

- Shows logs from a specific timeframe.

```
Apr 30 15:21:05 cis106 systemd[829]: Queued start job for default target default.target.
Apr 30 15:21:05 cis106 systemd[829]: Created slice app.slice - User Application Slice.
Apr 30 15:21:05 cis106 systemd[829]: Reached target paths.target - Paths.
Apr 30 15:21:05 cis106 systemd[829]: Reached target timers.target - Timers.
Apr 30 15:21:05 cis106 systemd[829]: Starting dbus.socket - D-Bus User Message Bus Socket...
Apr 30 15:21:05 cis106 systemd[829]: Starting ssh-agent.socket - OpenSSH Agent socket...
Apr 30 15:21:05 cis106 systemd[829]: Listening on ssh-agent.socket - OpenSSH Agent socket.
Apr 30 15:21:05 cis106 systemd[829]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Apr 30 15:21:05 cis106 systemd[829]: Reached target sockets.target - Sockets.
Apr 30 15:21:05 cis106 systemd[829]: Reached target basic.target - Basic System.
Apr 30 15:21:05 cis106 systemd[829]: Started mpris-proxy.service - Bluetooth mpris proxy.
Apr 30 15:21:05 cis106 systemd[829]: Reached target default.target - Main User Target.
Apr 30 15:21:05 cis106 systemd[829]: Startup finished in 154ms.
Apr 30 15:21:05 cis106 systemd[829]: Created slice session.slice - User Core Session Slice.
Apr 30 15:21:05 cis106 systemd[829]: Starting dbus.service - D-Bus User Message Bus...
Apr 30 15:21:05 cis106 systemd[829]: Started dbus.service - D-Bus User Message Bus.
Apr 30 15:27:45 cis106 sudo[907]: ilovetrains28 : TTY=ttty1 ; PWD=/home/ilovetrains28 ; USER=root ; COMMAND=/usr/sbin/apache2ctl -M
Apr 30 15:27:45 cis106 sudo[907]: pam_unix(sudo:session): session opened for user root(uid=0) by ilovetrains28(uid=1000)
Apr 30 15:27:45 cis106 sudo[907]: pam_unix(sudo:session): session closed for user root
Apr 30 15:35:46 cis106 sudo[951]: ilovetrains28 : TTY=ttty1 ; PWD=/home/ilovetrains28 ; USER=root ; COMMAND=/usr/sbin/azensite site.conf
Apr 30 15:35:46 cis106 sudo[951]: pam_unix(sudo:session): session opened for user root(uid=0) by ilovetrains28(uid=1000)
Apr 30 15:35:46 cis106 sudo[951]: pam_unix(sudo:session): session closed for user root
Apr 30 15:40:49 cis106 sudo[983]: ilovetrains28 : TTY=ttty1 ; PWD=/home/ilovetrains28 ; USER=root ; COMMAND=/usr/sbin/apache2ctl -S
Apr 30 15:40:49 cis106 sudo[983]: pam_unix(sudo:session): session opened for user root(uid=0) by ilovetrains28(uid=1000)
Apr 30 15:40:49 cis106 sudo[983]: pam_unix(sudo:session): session closed for user root
Apr 30 15:44:56 cis106 sudo[995]: ilovetrains28 : TTY=ttty1 ; PWD=/home/ilovetrains28 ; USER=root ; COMMAND=/usr/bin/vim /etc/apache2/ports.conf
Apr 30 15:44:56 cis106 sudo[995]: pam_unix(sudo:session): session opened for user root(uid=0) by ilovetrains28(uid=1000)
-- Boot 4347f6d6c55143c2af99dfc4d549af57 --
Apr 30 15:47:12 cis106 systemd[820]: Queued start job for default target default.target.
Apr 30 15:47:12 cis106 systemd[820]: Created slice app.slice - User Application Slice.
Apr 30 15:47:12 cis106 systemd[820]: Reached target paths.target - Paths.
Apr 30 15:47:12 cis106 systemd[820]: Reached target timers.target - Timers.
Apr 30 15:47:12 cis106 systemd[820]: Starting dbus.socket - D-Bus User Message Bus Socket...
Apr 30 15:47:12 cis106 systemd[820]: Starting ssh-agent.socket - OpenSSH Agent socket...
Apr 30 15:47:12 cis106 systemd[820]: Listening on ssh-agent.socket - OpenSSH Agent socket.
Apr 30 15:47:12 cis106 systemd[820]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Apr 30 15:47:12 cis106 systemd[820]: Reached target sockets.target - Sockets.
Apr 30 15:47:12 cis106 systemd[820]: Reached target basic.target - Basic System.
Apr 30 15:47:12 cis106 systemd[820]: Started mpris-proxy.service - Bluetooth mpris proxy.
Apr 30 15:47:12 cis106 systemd[820]: Reached target default.target - Main User Target.
Apr 30 15:47:12 cis106 systemd[820]: Startup finished in 131ms.
Apr 30 15:47:12 cis106 systemd[820]: Created slice session.slice - User Core Session Slice.
Apr 30 15:47:12 cis106 systemd[820]: Starting dbus.service - D-Bus User Message Bus...
Apr 30 15:47:12 cis106 systemd[820]: Started dbus.service - D-Bus User Message Bus.
Apr 30 15:48:20 cis106 sudo[839]: ilovetrains28 : TTY=ttty1 ; PWD=/home/ilovetrains28 ; USER=root ; COMMAND=/usr/bin/vim /etc/apache2/apache2.conf
Apr 30 15:48:20 cis106 sudo[839]: pam_unix(sudo:session): session opened for user root(uid=0) by ilovetrains28(uid=1000)
-- Boot c0666176c7ae424796713d7aca003937 --
Apr 30 15:59:10 cis106 systemd[824]: Queued start job for default target default.target.
Apr 30 15:59:10 cis106 systemd[824]: Created slice app.slice - User Application Slice.
```

Activate